
Online Examination Report

Date:	28.07.2014
Attempts:	1
Examinee:	Kibuuka, Elvis
Date of birth:	11/16/1989
Subject:	080- Principles of flight
Result:	0 %
Time used:	00:00:11 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	226	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
2	148	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
3	201	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
4	232	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
5	188	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
6	240	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
7	197	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
8	189	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
9	209	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
10	206	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
11	207	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
12	195	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
13	231	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
14	192	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
15	214	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
16	113	81.1	SUBSONIC AERODYNAMICS	0,00	1,00
17	48	81.1.1.1	Laws and definitions	0,00	1,00
18	84	81.1.3.1	The lift coefficient C_l	0,00	1,00
19	57	81.1.4.2	Induced drag	0,00	1,00
20	4	81.1.4.2	Induced drag	0,00	1,00
21	35	81.1.6	Ground effect	0,00	1,00
22	38	81.1.8.2	The stall speed	0,00	1,00
23	44	81.1.8.5	Special phenomena of stall	0,00	1,00
24	26	81.1.10.1	Spoilers and the reasons for use	0,00	1,00
25	52	81.2.2.1	Normal shock waves	0,00	1,00
26	16	81.2.2.2	Oblique shock waves	0,00	1,00
27	45	81.2.4	Buffet onset	0,00	1,00
28	104	81.4	STABILITY	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	79	81.4.3.7	Factors affecting the C_m - α graph	0,00	1,00
30	10	81.5.8.3	Stabiliser trim	0,00	1,00
31	96	81.6.1.1	Flutter	0,00	1,00
32	109	81.7	PROPELLERS	0,00	1,00
33	50	81.7.1.3	Fixed and variable pitch/constant speed	0,00	1,00
34	51	81.7.1.4	Propeller efficiency versus speed	0,00	1,00
35	73	81.7.1.4	Propeller efficiency versus speed	0,00	1,00
36	83	81.7.4.2	Gyroscopic precession	0,00	1,00
37	89	81.8.1.3	Straight steady descent	0,00	1,00
38	29	81.8.1.4	Straight steady glide	0,00	1,00
39	31	81.8.1.4	Straight steady glide	0,00	1,00
40	92	81.8.3	Particular points on the polar curve	0,00	1,00
Total:0%				0,00	40,00

Attachment 2: List of result by topic

Licence

Date: 28.07.2014

Examinee: Kibuuka, Elvis

Location: Uganda Civil Aviation Authority

Your Result: 0,00 from 40,00 Points (0,00 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	0,00	40,00	0,00
81.1. SUBSONIC AERODYNAMICS	9	0,00	9,00	0,00
81.2. HIGH SPEED AERODYNAMICS	3	0,00	3,00	0,00
81.4. STABILITY	2	0,00	2,00	0,00
81.5. CONTROL	1	0,00	1,00	0,00
81.6. LIMITATIONS	1	0,00	1,00	0,00
81.7. PROPELLERS	5	0,00	5,00	0,00
81.8. FLIGHT MECHANICS	4	0,00	4,00	0,00
Total	40	0,00	40,00	0,00

Candidate: Elvis Kibuuka
Date of birth: 1989-11-16

Subject: 080- Principles of flight
Date: 2014-07-28



1

During a wing stall, the wing tips of a weight shift aircraft are

- ☐ ineffective for stall recovery.
- ☐ effective only when combined with maximum engine output.
- ☒ effective for stall recovery.

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2

When a glider is turning in flight, the force that opposes the inward turning force is called

☐ resultant force.

☐ adverse yaw.

☒ centrifugal force.

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3

A rapid acceleration can create the illusion of being in a

☐ left turn.

☐ nosedown attitude.

☒ noseup attitude.

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4

During a 'no-gyro' approach and prior to being handed off to the final approach controller, the pilot should make all turns

☐ one-half standard rate unless otherwise advised.

☒ standard rate unless otherwise advised.

☐ any rate not exceeding a 30 degrees bank.

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5

In a twin-engine airplane, the single-engine service ceiling is the maximum density altitude at which V(YSE) will produce

- ☐ 100 feet per minute rate of climb.
- ☒ 50 feet per minute rate of climb.
- ☐ 500 feet per minute rate of climb.

6

(Refer to Figure 146.) Identify the system that has failed and determine a corrective action to return the airplane to straight-and-level flight.

- ☐ Vacuum system has failed; reduce power, roll left to level wings, and pitchup to reduce airspeed.
- ☒ Static/pitot system is blocked; lower the nose and level the wings to level-flight attitude by use of attitude indicator.
- ☐ Electrical system has failed; reduce power, roll left to level wings, and raise the nose to reduce airspeed.

See Attachments:

1. instrument_146.jpg

figure 146

(Attachment
No.1)

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7

If the gyroplane's CG is below the propeller thrust line, which direction will the application of power cause the nose to move?

- ☐ The nose will not move.
- ☒ The nose will pitch down.
- ☐ The nose will pitch up.

8

When one engine fails on a twin-engine airplane, the resulting performance loss

- ☐ reduces cruise indicated airspeed by 50 percent or more.
- ☐ is approximately 50 percent since 50 percent of the normally available thrust is lost.
- ☒ may reduce the rate of climb by 80 percent or more.

9

Unless adequate speed control is maintained during the turn to base and the final approach for a landing into the wind, which would most likely occur if a steep wind gradient existed?

- ☐ The airspeed on final approach would increase, causing the glider to overshoot the desired landing spot.
- ☒ The desired landing spot would be undershot or the glider would stall.
- ☐ The wingtip on the outside of the turn would stall before the wingtip on the inside of the turn.

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10

During an autolaunch, the pitch angle of the glider should not exceed

- ☐ 10 degrees at 50 feet, 20 degrees at 100 feet, and 45 degrees at 200 feet.
- ☒ 15 degrees at 50 feet, 30 degrees at 100 feet, and 45 degrees at 200 feet.
- ☐ 15 degrees at 50 feet, 20 degrees at 100 feet, and 40 degrees at 200 feet.

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11

At what point during an autotow should the glider pilot establish the maximum pitch attitude for the climb?

☐ Between 300 and 400 feet above the ground.

☐ 100 feet above the ground.

☒ 200 feet above the ground.

12

When an aircraft's forward CG limit is exceeded, it will affect the flight characteristics of the aircraft by producing

- ☒ **higher stalling speeds and more longitudinal stability.**
- ☐ **very light elevator control forces which make it easy to inadvertently overstress the aircraft.**
- ☐ **improved performance since it reduces the induced drag.**

13

If both the ram air input and drain hole of the pitot system are blocked, what airspeed indication can be expected?

- ☐ Decrease of indicated airspeed during a climb.
- ☐ Constant indicated airspeed during a descent.
- ☒ No variation of indicated airspeed in level flight even if large power changes are made.

14

An aircraft is loaded with the CG aft of the aft limit. What effect will this have on controllability?

- ☐ A stall will occur at a higher indicated airspeed due to the greater downloading on the elevator.
- ☐ A stall will occur at a lower airspeed, but recovery will be easier because of reduced wing loading.
- ☒ Stall and spin recovery may be difficult or impossible.

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What is the suggested speed to fly when passing through lift with no intention to work the lift?

☒ **Minimum sink speed.**

☐ **Best glide speed.**

☐ **Best lift/drag speed.**

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16

When the angle of attack of a symmetrical airfoil is increased, the center of pressure will

- ☒ have very little movement.
- ☐ remain unaffected.
- ☐ move aft along the airfoil surface.

17

Considering subsonic incompressible airflow through a Venturi, which statement is correct?

- I. The static pressure in the undisturbed airflow is lower than in the throat.
II. The speed in the undisturbed airflow is lower than in the throat.

- ☐ I is correct, II is correct.
- ☐ I is correct, II is incorrect.
- ☒ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.

18

The lift formula can be written as:
(rho = density)

☒ $L = C_L * \frac{1}{2} \rho * V^2 * S.$

☐ $L = C_L * 2 \rho * V^2 * S.$

☐ $L = n * W.$

☐ $L = W.$

19

The induced drag:

- ☐ increases as the aspect ratio increases.
- ☐ increases as the magnitude of the tip vortices decreases.
- ☒ increases as the lift coefficient increases.
- ☐ has no relation to the lift coefficient.

20

Which statement concerning the local flow pattern around a wing is correct?

- ☒ By fitting winglets to the wing tip, reduces induced drag.
- ☐ Sweepback reduces drag since, because the span increases.
- ☐ Vortex generators on the wing partially block the spanwise flow over the wing
- ☐ Slat extension, at a constant angle of attack and normal extension speeds, will increase the lift coefficient,

21

Which statement, about an aeroplane leaving ground effect at constant angle of attack, is correct?

- I. The lift coefficient C_L remains constant.
II. The induced drag coefficient C_{Di} decreases.

- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is correct.
- ☐ I is correct, II is incorrect.
- ☒ I is incorrect, II is incorrect.

22

The stall speed increases, when: (all other factors of importance being constant)

☐ minor altitude changes occur e.g. 0-10.000 ft.

☐ weight decreases.

☒ pulling out of a dive.

☐ spoilers are retracted.

23

Regarding deep stall characteristics, identify whether the following statements are correct or incorrect:

I. The combination of a wing with sweepback and a T-tail make an aeroplane prone to deep stall.

II. A stick pusher system can be fitted to an aeroplane that exhibits abnormal stall characteristics.

☐ I is correct, II is incorrect.

☐ I is incorrect, II is correct.

☒ I is correct, II is correct.

☐ I is incorrect, II is incorrect.

24

When a wing spoiler is extended at constant angle of attack:

- ☐ both drag and lift increase.
- ☐ both drag and lift decrease.
- ☐ drag increases but lift remains constant.
- ☒ drag increases but lift decreases.

25

**Which of the following statements is correct?
Air passes a normal shock wave as**

- ☐ The static pressure decreases.
- ☐ The velocity increases.
- ☒ The static temperature increases.
- ☐ The static temperature decreases.

26

Compared with an oblique shock wave a normal shock wave has a:

- ☐ lower static temperature.
- ☒ higher loss in total pressure.
- ☐ higher total temperature.
- ☐ higher total pressure.

27

What is the significance of the maximum allowed cruising altitude, based on the 1.3 g margin?

- ☒ a manoeuvre with a load factor of 1.3 will cause buffet onset.
- ☐ a manoeuvre with a load factor of 1.3 will cause a Mach number at which accelerated low speed stall occurs.
- ☐ exceeding a load factor of 1.3 will cause permanent deformation of this aeroplane.
- ☐ a manoeuvre with a load factor of 1.3 will cause Mcrit to be exceeded.

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28

If an aircraft has negative dynamic and positive static stability, this will result in

- ☒ divergent oscillations.
- ☐ convergent oscillations.
- ☐ undamped oscillations.

29

(Please view annex 081-0027 issue date July 2006)

Which of these statements about the pitching moment coefficient versus angle of attack lines in the annex is correct?

- ☒ The CG position is further aft at line 1 when compared with line 4.
- ☐ In its curved part at high angles of attack line 2 illustrates a decreasing static longitudinal stability.
- ☐ Static longitudinal stability is greater at line 3 when compared with line 4 at low and moderate angles of attack.
- ☐ The CG position is further forward at line 1 when compared with line 4.

See Attachments:

1. 081-0027.gif

Annex

(Attachment
No.2)

30

Comparing the differences between a horizontal trimmable stabiliser and an elevator trim tab, which statement is correct?

- I. A stabiliser trim is less suitable for jet transport aeroplanes because of their large speed range.
II. A trim tab is a more powerful means of trimming.

☒ I is incorrect, II is incorrect.

☐ I is correct, II is correct.

☐ I is incorrect, II is correct.

☐ I is correct, II is incorrect.

31

Wing flutter may be caused by a:

- ☐ aerodynamic wing stall at high speed.
- ☒ combination of bending and torsion of the wing structure.
- ☐ combination roll control reversal and low speed stall.
- ☐ combination of fuselage bending and wing torsion.

32

A propeller rotating clockwise, as seen from the rear, creates a spiraling slipstream that tends to rotate the aircraft to the

- ☒ left around the vertical axis, and to the right around the longitudinal axis.
- ☐ left around the vertical axis, and to the left around the longitudinal axis.
- ☐ right around the vertical axis, and to the left around the longitudinal axis.

33

The angle of attack of a fixed pitch propeller blade increases when:

- ☐ velocity and RPM increase.
- ☐ velocity and RPM decrease.
- ☐ forward velocity increases and RPM decreases.
- ☒ RPM increases and forward velocity decreases.

34

Which statement is correct when comparing a fixed pitch propeller with a constant speed propeller?

- I. A fixed pitch propeller improves propeller efficiency over a range of cruise speeds.
II. A coarse fixed pitch propeller is more efficient during take-off.

- ☐ I is correct, II is correct.
- ☒ I is incorrect, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is incorrect.

35

Which statement is correct when comparing a fixed pitch propeller with a constant speed propeller?

- I. A constant speed propeller reduces fuel consumption over a range of cruise speeds.
II. A coarse fixed pitch propeller is more efficient during take-off.

- ☐ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☒ I is correct, II is incorrect
- ☐ I is correct, II is correct.

36

Which statement is correct regarding the gyroscopic effect of a clockwise rotating propeller on a single engine aeroplane?

- I. Pitch down produces left yaw.
II. Left yaw produces pitch down.

- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is correct.
- ☒ I is correct, II is incorrect.
- ☐ I is incorrect, II is incorrect.

37

During a straight steady descent, lift is:

- ☒ less than weight,
- ☐ greater than weight because it must additionally balance a component of drag.
- ☐ equal to weight to maintain a constant speed.
- ☐ less than weight

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38

Which of the following increases the maximum duration of a glide?

- ☒ A decrease in mass.
- ☐ A headwind.
- ☐ A tailwind.
- ☐ An increase in mass.

39

What factors determine the distance travelled over ground of an aeroplane in a glide from a given altitude?

- ☐ The wind and weight together with power loading, which is the ratio of power output to the weight.
- ☒ The wind and the lift/drag ratio.
- ☐ The wind and CLMAX.
- ☐ The wind and the aeroplane's mass.

40

(Please view annex 081-0007 issue date October 2005)
The point in the diagram giving the lowest speed in unaccelerated flight is:

☐ point 1.

☐ point 3.

☒ point 4.

☐ point 2.

See Attachments:

1. 081-0007.gif

Annex

(Attachment
No.3)

Attachment No. 1

Questions 6

figure 146

instrument_146.jpg

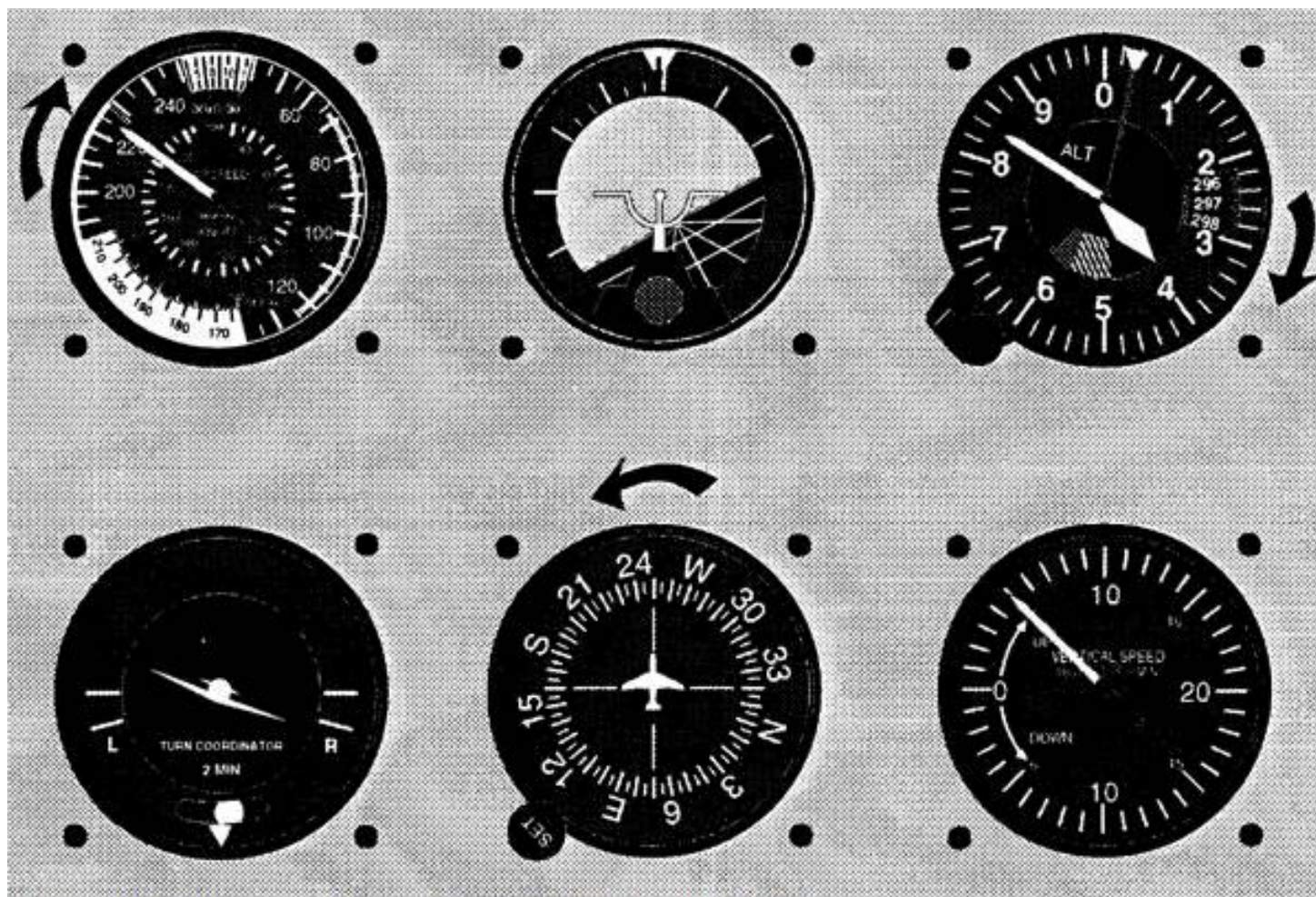


Figure 146. Instrument Sequence (System Failed) © ASA

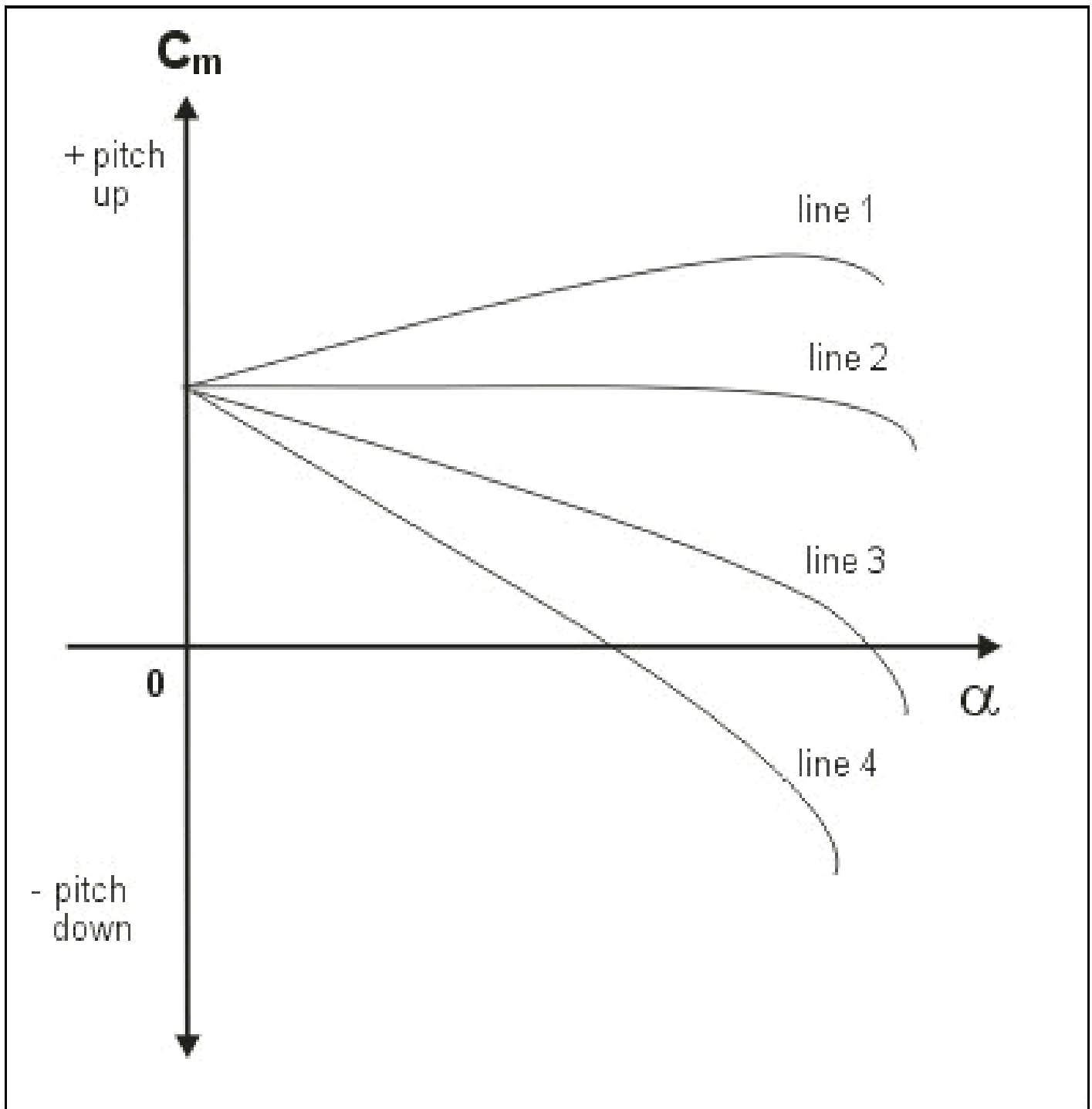
Attachment No. 2

Questions 29

Annex

081-0027.gif

081-0027 issue date July 2006



Attachment No. 3

Questions 40

Annex

081-0007.gif

081 - 0007

